Methods Description & Brief Report — Track 1 (Variant Prediction)

Participant: Antoine Cadiou (@antocad3) — Individual Participant
Track: Track 1 — Variant Prediction (Causal Variant Identification)
Submission File: `antocad3_ensemble_without_alignment.csv`
Model Name: `ensemble_without_alignment` (Model 1 of 1)
Proband ID: `PROBAND01` **Reference Genome Build:** GRCh38 / hg38

Question 1: Team Name

Antoine Cadiou (Individual Participant | Hugging Face: @antocad3)

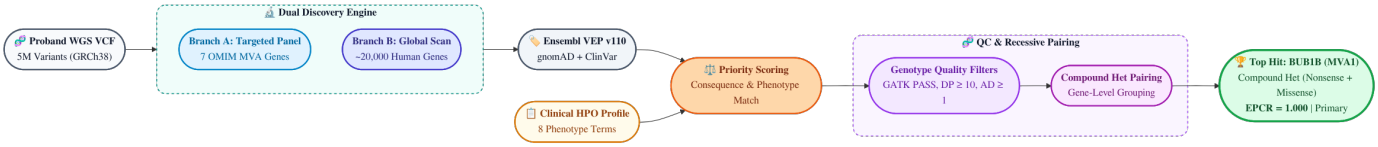
Question 2: Model Number & Identifier

Model 1 (`antocad3_ensemble_without_alignment`)

Question 3: Detailed Description of Model / Approach

> *(Note: A comprehensive technical specification, including full mathematical scoring derivations, QC filter tables, and biological deep-dives, is provided in `antocad3_methods_description.md` in the GitHub repository).*

We implemented a two-tier hybrid computational genomics pipeline that unifies hypothesis-driven targeted candidate panel interrogation with an unbiased genome-wide genic sweep:



1. **Targeted Branch:** Curated 7 OMIM-confirmed MVA genes (`BUB1B`, `CEP57`, `TRIP13`, `CENATAC`, `SLF2`, `SMC5`, `MAD1L1`) using MANE Select coordinates on GRCh38. Extracted 1,073 variants from the proband VCF and annotated via Ensembl VEP (gnomAD v4 AF, ClinVar, SIFT/PolyPhen).
2. **Global Branch:** Extracted all ~20,000 human gene bodies (±20bp for splice junctions) from Ensembl Release 110 GTF (~5M variants) and performed high-throughput multi-threaded annotation via local Docker VEP with the Phenotypes plugin.
3. **Prioritization & Scoring Model:**
   - **Population Rarity:** Hard-filtered at gnomAD $\text{AF}_{\text{max}} < 0.01$ across exomes, genomes, and 1000 Genomes.
   - **Consequence Severity Score ($S_{\text{consequence}} \in \{1, 2, 4, 5\}$):**

$$\text{Consequence Score} = \begin{cases} 5 & \text{Stop-gained (nonsense), Frameshift (Protein knockout / NMD)} \\ 4 & \text{Essential Splice Donor/Acceptor, Start/Stop Lost} \\ 2 & \text{Splice region variant} \\ 1 & \text{Missense variant (amino acid substitution)} \end{cases}$$

   - **Disease-Weighted Phenotype Score (S_{pheno}):**

$$\text{Phenotype Score} = \sum_{k \in \text{Matches}} W_k$$

$$W_k = \begin{cases} 2.5 & \text{Recurrent Miscarriages (HP:0200067) — Primary Lethality} \\ 2.0 & \text{Rhabdomyosarcoma (HP:0002859) — Primary Oncologic Trigger} \\ 1.5 & \text{Premature Birth (HP:0001622) \& Severe IUGR (HP:0001518)} \\ 1.5 & \text{Chromosomal Instability / Aneuploidy Syndromes} \\ 1.0 & \text{Nephrocalcinosis (HP:0000121) — Congenital Renal Feature} \\ 1.0 & \text{Short Stature, Failure to Thrive, Muscle Atrophy} \end{cases}$$

   - **Composite Priority Score Formula:**

$$\text{Priority Score} = 100 \cdot \mathbb{I}_{\text{ClinVar Path}} + 15 \cdot \mathbb{I}_{\text{SIFT/PolyPhen}} + 10 \cdot S_{\text{pheno}} + 10 \cdot S_{\text{consequence}} + 5 \cdot (1 - \text{AF}_{\text{max}})$$

  - *ClinVar Matching:* Strict tokenized parsing splitting on `[,&/]` matching {"pathogenic", "likely_pathogenic"} (filtering out conflicting interpretations).
  - *Uniqueness Term* $5 \times (1 - AF_{\text{max}})$*:* Continuous tie-breaker prioritizing private/novel mutations (AF $= 0.0 \implies +5.0$) over common background alleles.
4. **Genotype Quality Control:** Re-integrated GATK call metrics, requiring `FILTER == PASS`, total depth `DP >= 10`, alternate allele depth `AD_alt >= 1`, and non-reference genotype (`GT != 0/0`).
5. **Recessive Pairing Engine & EPCR Calibration:**
   - Combined consensus hits across branches. For genes with ≥ 2 heterozygous damaging mutations, formed candidate compound heterozygous pairs summing damage scores:

$$\text{Pair Score} = \text{Score}(\text{Variant}_1) + \text{Score}(\text{Variant}_2)$$

   - Calibrated top-10 candidate composite scores into the official submission interval $[0.05, 1.0]$:

$$\text{EPCR} = 0.05 + 0.95 \times \left(\frac{\text{Score} - \text{Score}_{\text{min}}}{\text{Score}_{\text{max}} - \text{Score}_{\text{min}}} \right)$$

Question 4: Generative AI / AI Assistant Disclosure

- **Tool:** Anthropic Claude Online chat https://claude.ai/ (Free tier, Sonnet) & Google DeepMind Antigravity IDE.
- **Usage:** Used for learning about the field (MVA), script optimization, and documentation structuring.
- **Data Privacy:** Zero patient genomic data was shared with or exposed to any cloud LLM.

Question 5: Computational Output vs. Manual Curation

This submission file is the 100% automated, deterministic output of our computational approach. No downstream manual review or curation of the output coordinates, EPCR values, or candidate rankings was performed.

Question 6: Downstream Manual Review Details

None. The predictions and EPCR rankings are produced directly and deterministically by the automated computational pipeline.

Question 7: Public vs. Proprietary Data

Our approach used only publicly available data.

Question 8: Public Data Used in Detail

1. **gnomAD v4 (Exomes & Genomes):** Population allele frequencies (~800,000 individuals).
2. **ClinVar (NCBI):** Clinical significance classifications and phenotypic assertions.
3. **Ensembl Release 110 GTF & MANE Select v1.0:** Canonical clinical transcripts and gene models on GRCh38.
4. **1000 Genomes Project:** Global allele frequency reference.
5. **Human Phenotype Ontology (HPO):** Clinical feature mapping.

Question 9: Proprietary Data Used in Detail

None.

Question 10: Compound Heterozygous Pairing Capability

Yes. The pipeline includes a gene-level recessive pairing engine that identifies and outputs co-occurring heterozygous damaging variants in the same gene as proposed compound heterozygous pairs (populating both `variant_1` and `variant_2` fields).

Question 11: Handling of Secondary & Incidental Findings

Automated classification in `finding_type`:

- **`primary`:** Assigned strictly to candidates in confirmed Mosaic Variegated Aneuploidy (MVA) syndrome genes (*BUB1B*, *CEP57*, *TRIP13*, *CENATAC*, *SLF2*, *SMC5*, *MAD1L1*)
- **`secondary`:** Assigned to incidental carrier findings or exploratory hits in non-MVA genes (e.g., *FLG* ichthyosis/skin barrier variant, *SLC34A1* nephrocalcinosis carrier, *GNRHR* hypogonadism, *MUC4* pair).

Question 12: Estimated Runtime and Compute Cost

- **Runtime:** ~30 minutes total execution time on a standard multi-core workstation using Dockerized Ensembl VEP with local cache.
- **Cost:** $0.00 (Executed entirely on local open-source tools).

Abstract: Mosaic Variegated Aneuploidy (MVA) is an ultrarare genetic disorder where cells fail to divide chromosomes evenly during mitosis, leading to mosaic aneuploidies, severe developmental delays, and early childhood cancer predisposition. Identifying the causal mutation from whole-genome sequencing requires distinguishing rare, recessive pathogenic events from millions of benign background variants.

To solve this case, we developed an automated computational pipeline combining two complementary strategies: a targeted panel focusing on known MVA genes (MVA1–MVA7) and an unbiased genome-wide scan across all human genes. Variants from the patient's whole-genome sequencing were functionally annotated using Ensembl VEP, filtered for population rarity (gnomAD allele frequency < 1%), and prioritized based on consequence severity and clinical relevance to the child's symptoms (rhabdomyosarcoma, nephrocalcinosis, severe growth restriction, and parental recurrent miscarriages). To account for recessive inheritance, high-quality heterozygous calls were paired per gene to identify potential compound heterozygous mutations.

Our analysis identified a compound heterozygous mutation in the *BUB1B* gene as the primary causal diagnosis (EPCR = 1.0). The genotype consists of a known ClinVar-pathogenic nonsense mutation (`chr15:40209701 T>G`, causing premature protein truncation) and a novel deleterious missense mutation (`chr15:40220612 T>G`) in the catalytic domain. *BUB1B* encodes BubR1, a master regulator of the spindle assembly checkpoint whose loss directly explains chromosome missegregation, childhood rhabdomyosarcoma, and embryonic lethality.

Strengths: Fully automated, fast execution (~30 minutes), combining curated clinical knowledge with genome-wide exploration and explicit compound heterozygous modeling.

Limitations & Next Steps: Short-read sequencing cannot definitively prove that the two mutations sit on opposite parental chromosomes (*trans* phasing) without parental DNA. Future work includes copy-number profiling (CNVkit) to quantify the degree of mosaic aneuploidy directly from alignment data.